

AdInsight: Comprehensive Analytics for Ad Performance and User Engagement Using Glance and Emotion Detection

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Chapter 1

Introduction

We aim to develop a software platform designed to enhance the analysis of video advertisements on web pages. The system will integrate directly with web browsers through a dedicated extension. This will enable it to detect ads, track user interactions on adverts, and gather metrics and data based on user engagement, all running on the host machine. We will be employing advanced technologies such as facial recognition, gaze detection, demographic analysis, and object detection so that the platform provides companies with comprehensive analytics to optimize their advertising strategies. In this document we aim to cover our motivation, problem statement and proposed solution for our project.[1]

1.1 Problem Statement

In this digital age, video advertisements are a main source for companies to showcase their products. Companies use them to gauge and gain user interest. However, in the current digital advertising landscape, advertisers lack critical insights into how users interact with ads like where users look, their level of engagement, and their emotional responses. This gap in understanding leads to missed opportunities for optimizing ad content, targeting the right audience, and improving ROI. These targets are critical for the success of advertisements. Moreover, the absence of detailed demographic data and sentiment analysis limits the ability of companies to make ads that are targeted for different user groups. Users too, do not have streamlined and relevant ad experiences which effects their overall interest and sentiments towards advertisements. To address these issues a more complex and privacy-compliant solution is needed which can track user interactions in real-time, extract meaningful data, and provide insights to improve advertising effectiveness without creating any privacy issues.[1]

1.2 Scope

The project aims to develop an advanced software platform that integrates with web pages through a browser extension. The software is designed to detect and analyze video advertisements, track user attention metrics by monitoring glances and engagement duration. It generates analytics of areas of interest, extracted objects of interest, utilizes demographic data, and emotion detection analysis to gauge user reactions and interest. The platform offers data driven analytics and meaningful insights for companies like ad companies, leading to better targeted adverts for users.[1]

1.3 Modules

1.3.1 Browser Extension + Host Machine Module

This module will run on the user's host machine as a browser extension. It will be responsible for:

- Detecting and Localizing Adverts.
- Tracking user Gaze.
- Tracking user Emotions and Demographics.

This module will locally process the captured frames of interest and send them to the server, which makes up the Server Module.

Features:

- **Ad Detection:** The browser extension will use EasyList to detect and localize video ads on the current web page so that their data can be corresponded with the gaze tracking part.
- **Gaze Detection:** The browser extension will use WebGazer.js to monitor and detect user gaze when the advert is being played.
- **Emotion and Demographic Analysis:** The browser extension will use DeepFace to detect both user emotions and demographics (age, gender) while viewing the advert.
- **Data Extraction:** The browser extension will extract frames of interest in the video ad, combine them with heatmaps generated from gaze detection, as well as with data from emotion and demographic analysis, and send frames to the server for further processing and object detection.

1.3.2 Server Module

This module will process the frames sent over by the browser extension, extract meaningful insights from them, and then deliver these insights to the analytics dashboard.

Features:

- **Manage Advert Information:** The browser extension will send ad metadata, which the server will use to extract the video and related data. It will then use Object Detection to extract objects of interest in the video and keep their record.
- **Object Detection:** The object detection part of the server will parse over advert videos and extract objects of interest and store them. The extracted objects will later be used with the processed frames from the browser extension to extract key objects of interest and other useful analytics.
- **Processing Data:** The processed data from the browser extension will be combined with the extracted objects to produce the final processed output. Meaningful analytics will be extracted from this output, which will be stored and can be accessed anytime by the analytics dashboard.

1.3.3 Analytics Frontend Module

This module will provide a GUI where advertising companies and stakeholders will be able to view the gathered analytics of their posted ads.

Features:

- **Ad Specific Analytics:** The ad-specific part will offer a detailed view of the ad performance of a specific ad. It will provide metrics on ad performance, such as users' glance duration, heatmaps, objects of interest from the user's perspective, and their emotions and demographics.
- **Report Generation:** Generate detailed reports on the performances of all ads posted by the advertising company. Provide an analysis of each ad and how well users are responding to it, and what is the common object of interest for most viewers.
- **User Demographics:** View all your ad viewers' demographics such as age and gender, as well as their emotional responses while watching the ads, allowing advertising companies to make content specifically tailored to their consumer market.

1.4 User Classes and Characteristics

User Class	Description
Advertising Companies/ Marketers	Primary users who are interested in ad performance and user engagement analytics. They require insights into user behavior, including gaze tracking, emotion detection, and demographic data. They may access data reports, heatmaps, and metrics via the analytics dashboard, focusing on optimizing ad campaigns and targeting specific demographics. Use Case: View detailed reports on ad performance, user interaction, and demographic information to make informed decisions about future ads.
Data Analysts	Skilled in analyzing large datasets and extracting meaningful insights. They likely utilize the detailed metrics generated from the system to identify trends and patterns, and are interested in optimizing data-driven ad strategies based on heatmaps, user emotions, and object detection data. Use Case: Interpret raw data from user interactions and convert it into actionable insights for stakeholders. Prepare detailed performance reports and identify key points of user engagement.
Ad Creators/ Content Designers	Individuals responsible for designing video ads based on the insights provided by the system. They are interested in how users engage with specific elements of the ad, such as objects or scenes, and use emotional and demographic feedback to tweak content for better performance. Use Case: Access user feedback and engagement data to design more engaging and effective ad content.
End Users/ Consumers	The individuals whose interactions are being tracked, such as their gaze, emotions, and demographic data. They may be concerned about privacy and security, thus requiring transparency about data collection practices. Their primary role is passive, as their engagement with ads provides the necessary data for analysis. Use Case: Interact with video ads during their normal browsing activity, with their behavior being analyzed without compromising privacy.

Table 1.1: User Profiles and Descriptions

Chapter 2

Project Requirements

2.1 Use-case/Event Response Table/Storyboarding

Event Type	Event	Trigger	Response	Activity
External	User installs the browser extension	User action: Browser extension installation	Initialize the extension and set up the necessary permissions	The extension prepares to start monitoring ad interactions.
External	Ad appears on the webpage	Video ad loaded on webpage	Detect the ad using EasyList and start tracking user interactions	The extension identifies a video ad and begins gathering metrics.
External	User views an ad	Video ad playback; user watching	Capture gaze data, emotional responses, and demographics	Monitor user interactions with the ad.
Internal	User gazes at the ad	WebGazer.js detecting user gaze	Log gaze coordinates and their duration	Capture where and for how long the user looks at the ad.
Internal	User's emotion detected	DeepFace analyzing user facial expressions	Record emotional data with corresponding timestamps	Analyze facial expressions to gauge user sentiment.
Internal	User demographic data captured	DeepFace processing user data	Store demographic insights associated with user interactions	Gather information on user demographics like age and gender.
Internal	Frame of interest extracted	WebGazer.js completing frame extraction	Combine gaze, emotion, and demographic data, then send to server	Prepare data for deeper analysis and object detection.
Internal	Data sent to server	Frame extraction completed	Store and process incoming data; trigger object detection	The server receives data and analyzes it.
Internal	Objects of interest detected	YOLO object detection triggered on frames	Extract and store key objects from the ad video	Identify specific elements in the ad that capture attention.
Internal	Processed data ready	Data processing complete	Generate reports and analytics for the frontend	Prepare insights for display in the analytics dashboard.
External	User accesses analytics dashboard	User action: Dashboard access	Retrieve and display relevant ad performance metrics	Provide companies with data-driven insights.
Internal	Reports generated	Request to generate analytics report	Provide detailed performance analysis on each ad	Summarize user engagement and emotional responses.

Figure 2.1: Event Response Table

2.2 Functional Requirements

2.2.1 Ad Detection

2.2.1.1 Detect Video Advertisements

- **Description:** The system shall detect and localize video advertisements on any web page the user visits.
- **Priority:** High
- **Details:**
 - The browser extension shall identify and localize video ads in real-time.
 - The system should detect video ads with at least 90% accuracy.
 - The system shall operate without impacting the user's regular browsing experience.

2.2.1.2 Correlate Ad Data with User Engagement

- **Description:** The system shall track user engagement with the identified advertisements.
- **Priority:** High
- **Details:**
 - Detected video ads should be linked to gaze and emotion tracking data collected during the ad's display.
 - The correlation should include timestamps and other metadata to ensure that user engagement is accurately tied to specific moments within the ad.

2.2.2 Gaze Detection

2.2.2.1 Monitor User Gaze

- **Description:** The system shall track the user's gaze to monitor engagement with the video ad.
- **Priority:** High
- **Details:**

- The browser extension shall track user eye movements.
- Gaze detection should start within 100 milliseconds after the video ad starts playing.
- The system shall track gaze at a frame rate of at least 30 FPS to ensure smooth detection of glances.

2.2.2.2 Generate Heatmaps Based on Gaze Data

- **Description:** The system shall generate heatmaps of the areas of interest on the screen based on user gaze data.
- **Priority:** Medium
- **Details:**
 - Heatmaps shall highlight the areas users focus on most within the ad.
 - The heatmaps should refresh every 500 milliseconds to ensure real-time insights.
 - Heatmaps shall be stored and sent to the server for further analysis.

2.2.3 Emotion and Demographic Analysis

2.2.3.1 Analyze User Emotions

- **Description:** The system shall analyze the emotions of users while watching video advertisements.
- **Priority:** High
- **Details:**
 - The browser extension shall detect user emotions during ad playback.
 - Emotion detection should occur in real-time and categorize emotions into happiness, sadness, surprise, anger, and neutral states.
 - The system should maintain an emotion recognition accuracy of at least 80%.

2.2.3.2 Capture Demographic Data

- **Description:** The system shall identify the demographic information (age group and gender) of users.

- **Priority:** Medium
- **Details:**
 - Categorize the age range (e.g., 18-24, 25-34) and gender of users.
 - Demographic data shall be inferred based on visual characteristics without violating user privacy.

2.2.4 Object Detection in Advertisements

2.2.4.1 Detect Objects in Video Advertisements

- **Description:** The system shall detect key objects in video advertisements that may capture user attention.
- **Priority:** High
- **Details:**
 - The server module shall detect objects within the ad frames sent by the browser extension.
 - Detected objects shall be stored with their respective timestamps for future correlation with gaze data.
 - The system should detect objects with an accuracy of at least 60%.

2.2.4.2 Identify Objects of Interest

- **Description:** The system shall correlate user gaze with detected objects to identify the objects that attracted the most attention.
- **Priority:** High
- **Details:**
 - Objects that users focus on for at least 2 seconds shall be flagged as "objects of interest."
 - These objects will be included in the ad performance report to help advertisers understand which ad elements captured user attention.

2.2.5 Data Storage and Processing

2.2.5.1 Store User Interaction Data

- **Description:** The system shall store all user engagement data, including gaze, emotional, and demographic data, in a secure database.
- **Priority:** Medium
- **Details:**
 - The browser extension shall send captured frames and data to the server for storage.
 - Data shall be stored in a manner compliant with privacy regulations, and the server shall ensure encryption at rest.

2.2.5.2 Generate Analytical Reports

- **Description:** The system shall process stored data to generate detailed analytical reports for advertisers.
- **Priority:** High
- **Details:**
 - The server module shall compile data on gaze tracking, emotion detection, demographics, and objects of interest into a report.
 - The reports shall include visual representations such as heatmaps and charts for ad performance.
 - Reports shall be exportable in multiple formats, including PDF and Excel.

2.3 Non-Functional Requirements

2.3.1 Performance

1. **Response Time:** The browser extension should detect and analyze an ad within 500 milliseconds of page load.
2. **Gaze Tracking Latency:** Gaze detection must have a latency of less than 100 milliseconds to ensure real-time tracking accuracy.

3. **Emotion Detection Speed:** The platform should detect and analyze user emotions from video frames within 500 milliseconds.
4. **Data Processing Time:** Server-side processing of frames and generation of insights should not exceed 5 seconds per video ad.

2.3.2 Scalability

1. **Simultaneous Users:** The platform must support concurrent users, handling multiple ad interactions and data processing.
2. **Ad Storage:** The server module should scale to store metadata and analysis of many video ads with minimal performance degradation.

2.3.3 Availability

1. **Uptime:** The system should maintain 99.95% uptime, ensuring that advertisers can access their analytics at any time.
2. **Fault Tolerance:** In case of server or network failures, the system should gracefully retry and process incomplete or delayed frames within 20 seconds.

2.3.4 Security

1. **Data Encryption:** All data exchanged between the browser extension and the server must be encrypted using at least AES-256 encryption.
2. **Privacy Compliance:** The system must comply with GDPR and similar privacy regulations, ensuring no personally identifiable information (PII) leaves the host machine.
3. **Authentication:** Multi-factor authentication should be implemented for users accessing the analytics dashboard.

2.3.5 Usability

1. **User Interface Response Time:** The analytics dashboard must render insights within 2 seconds after a user request.
2. **Browser Compatibility:** The extension should function on at least 95% of modern chromium web browsers (Chrome, Opera, and Edge) with a focus on responsive design.

2.3.6 Maintainability

1. **Modular Codebase:** The platform's codebase should maintain a modular architecture, ensuring that new features and bug fixes can be deployed within 2 hours.
2. **Documentation Coverage:** At least 90% of the codebase should have accompanying documentation to aid future development and maintenance.

2.3.7 Data Integrity

1. **Error Handling:** The system must gracefully handle errors in ad detection and data transmission, ensuring retries and logging within 5 seconds.

2.3.8 Interoperability

1. **Third-party Library Support:** The system must support seamless integration with libraries like WebGazer.js, EasyList, DeepFace, and YOLO without performance bottlenecks.

2.4 Domain Model

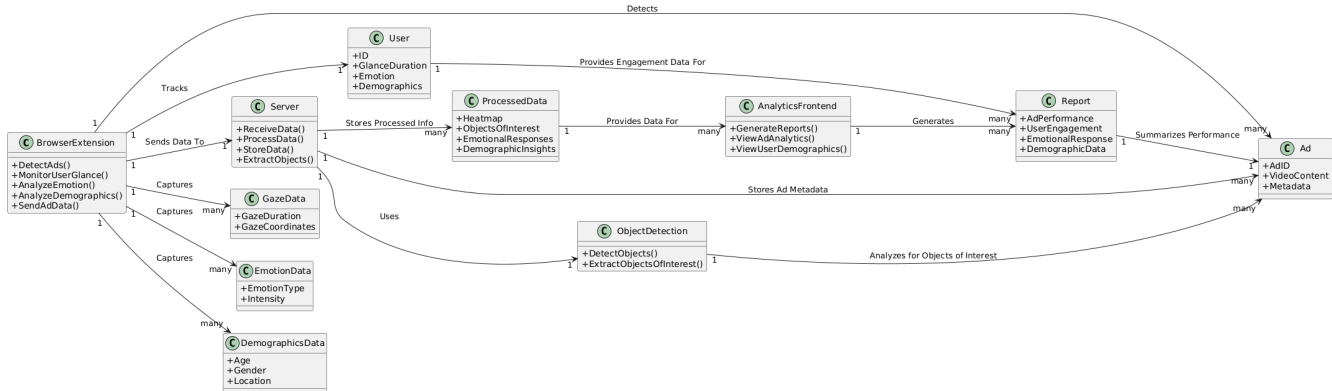


Figure 2.2: Domain Model

Chapter 3

System Overview

3.1 Architectural Design

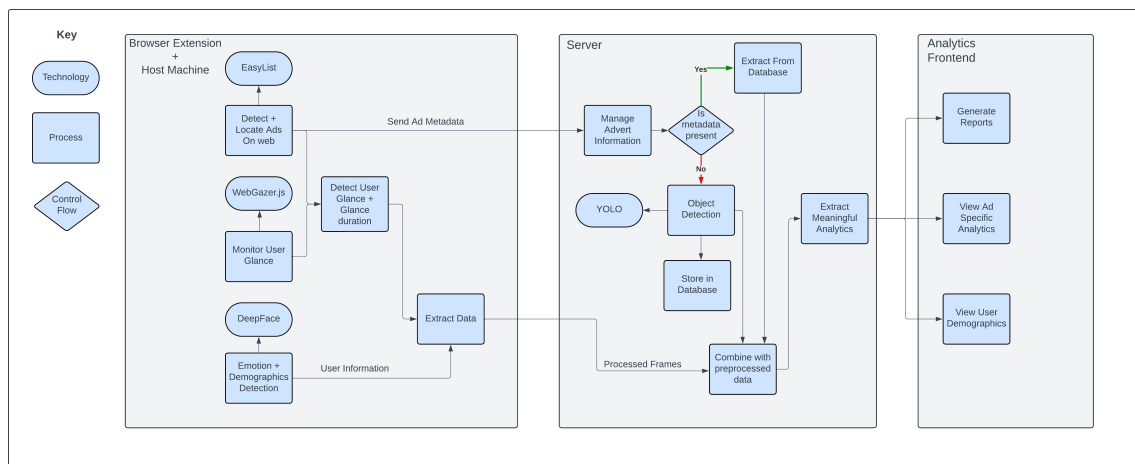


Figure 3.1: System Architecture

The architectural design diagram outlines the structure of our video advertisement analysis platform, detailing how various modules and technologies work together to deliver the desired functionality. It illustrates the flow of data, components, and processes from user interactions to the analytics provided to advertisers.

Browser Extension & Host Machine Module: This module operates locally on the user's device, detecting ads on web pages using EasyList. It tracks user gaze with WebGazer.js and captures emotions and demographic data with DeepFace. By processing video frames locally, we ensure that raw user data, especially facial images, is handled on the client side for privacy. The extension extracts key information like gaze duration, emotions, and demographics, securely transmitting it to the server for further processing.

Server Module: The server receives the processed data (frames, gaze tracking data, emotions, and demographics) from the browser extension. It employs object detection algorithms (like YOLO) to identify objects in video ads that may have captured user attention. The server combines user interactions with detected objects, generating insights about which elements attracted attention and how users emotionally responded. This processed data is then stored in the server-side databases for further analytics and reporting.

Analytics Dashboard: The dashboard serves as the interface for advertisers to view generated reports and insights. It pulls data from the server, offering detailed analytics, including heatmaps to indicate where users focused their gaze, emotional responses, demographics, and object detection results. Advertisers can filter and analyze this data to optimize future ad campaigns, enhancing engagement and targeting based on user behavior.

Data Flow: Data flows from user interactions with video ads to the browser extension, where gaze, emotions, and demographic data are captured and partially processed. This data is sent to the server for object detection and further analysis before being displayed on the analytics dashboard. Throughout the architecture, we prioritize privacy, ensuring sensitive data like facial information is processed locally on the user's device and not shared with the server.

3.2 Design Models

3.2.1 Data Flow Diagrams

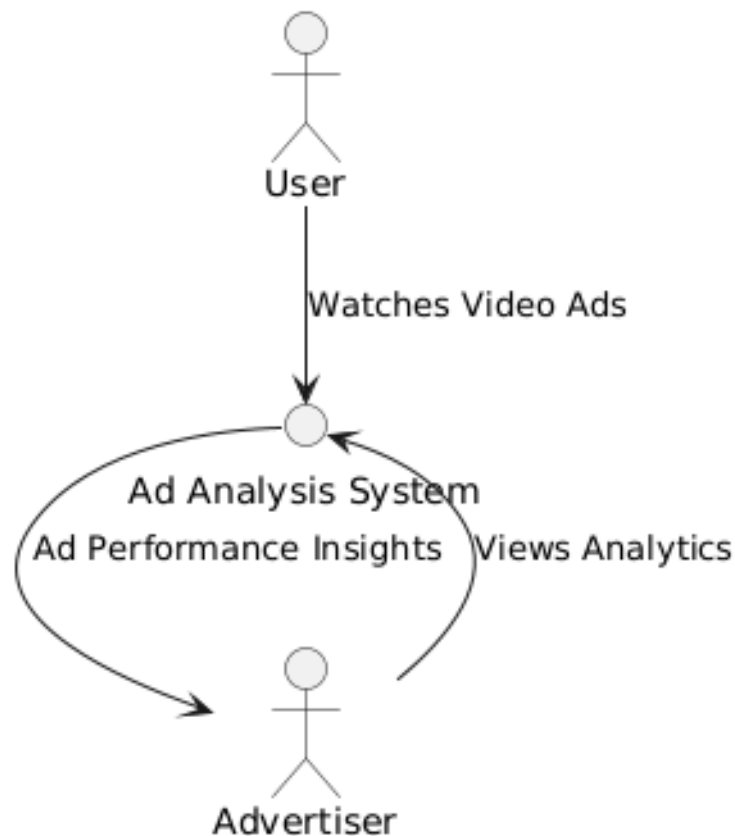


Figure 3.2: Level 0 Data Flow Diagram

DFD Level 0 (Context Diagram): This diagram provides a high-level overview of the system. It shows how external entities such as users and advertisers interact with the software. The main process, "Ad Analytics System," receives input from users via the browser extension and sends processed data to the advertisers for analysis, giving a simplified picture of how the system exchanges data.

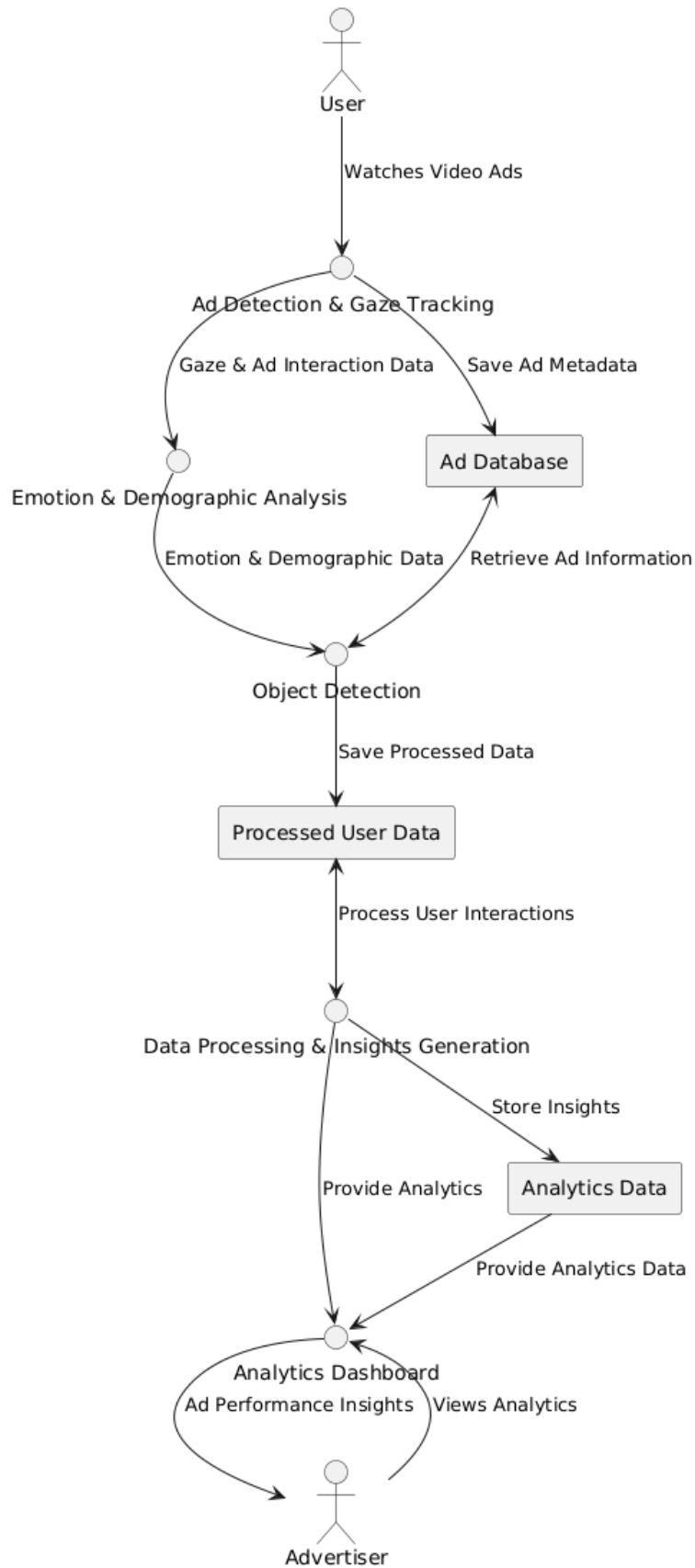


Figure 3.3: Level 1 Data Flow Diagram

DFD Level 1: This breaks the system into multiple sub-processes. The major processes include Ad Detection, Gaze Tracking, and Emotion/Demographic Analysis. Data flows between these processes and the browser extension, which collects raw data (user interactions with ads) and sends it to the server for further processing and analytics generation. This diagram offers a clearer picture of how the system operates at an intermediate level.

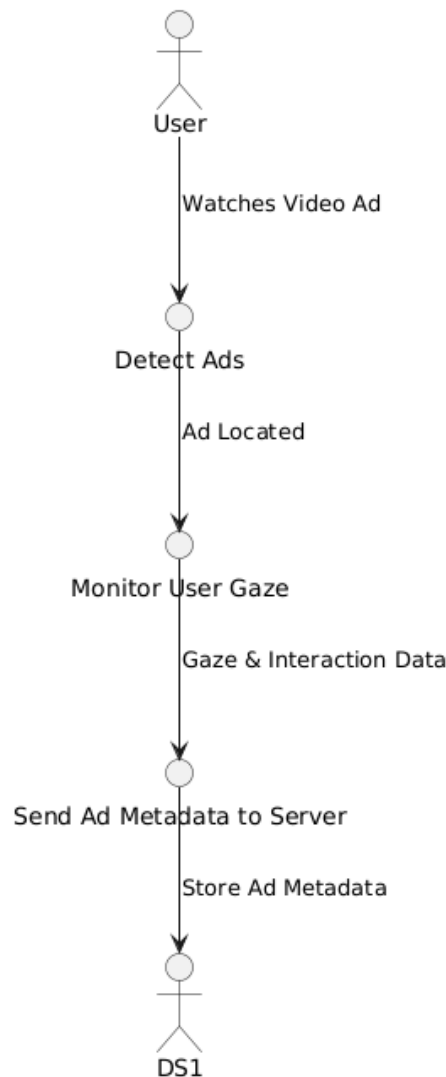


Figure 3.4: Level 2 Data Flow Diagram 1

DFD Level 2: DFD Level 2 expands on individual processes from DFD Level 1. For instance, the Ad Detection process is further broken down into specific actions like identifying video ads on a webpage using EasyList, while Emotion Analysis uses DeepFace to classify emotional responses. It provides more granularity and shows how data is handled within each module.

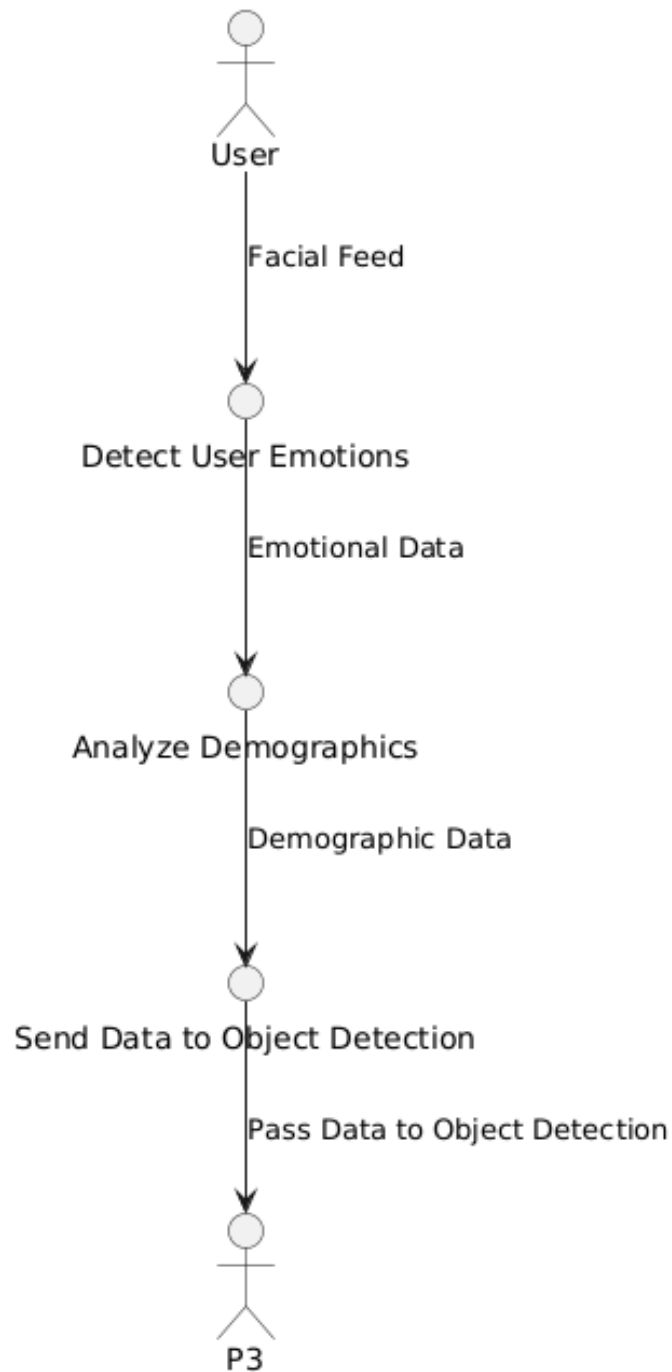


Figure 3.5: Level 2 Data Flow Diagram 2

This figure further expands on DFD Level 2 by focusing on how emotion and demographic analysis takes place, linking real-time data processing to the user's interactions with the ad content.

3.2.2 Activity Diagram

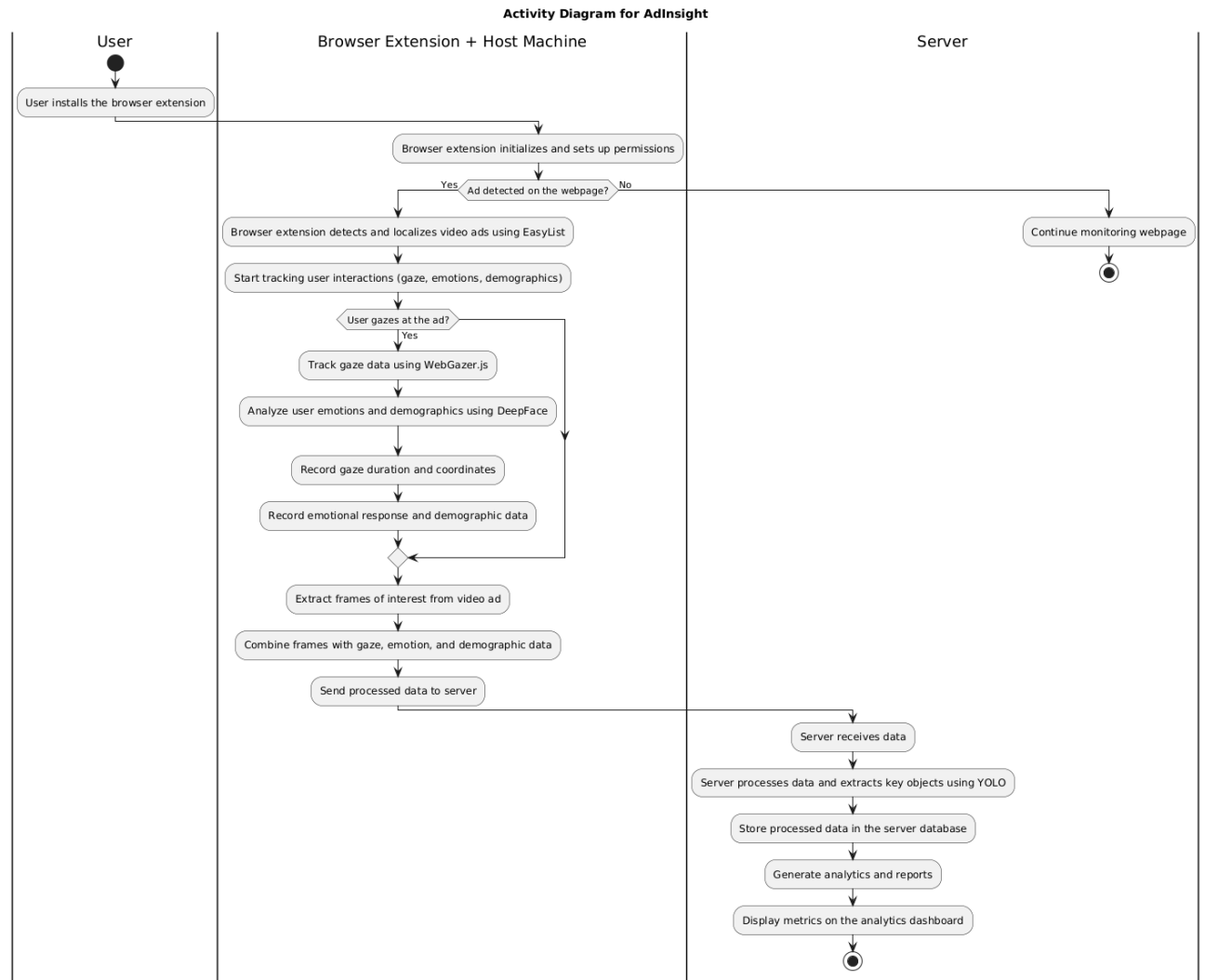


Figure 3.6: Activity Diagram

The activity diagram describes the flow of activities when users interact with the system. It details the sequence of actions, from ad detection by the browser extension to the collection of gaze and emotion data, followed by data processing on the server, and finally the generation of reports for advertisers.

3.2.3 System Sequence Diagram

The integrated System Sequence Diagram (SSD) illustrates the interactions between four key components: the *Web User*, *Browser Extension*, *Server Module*, and *Analytics Frontend*. The process begins when the *Web User* navigates a webpage, prompting the *Browser Extension* to monitor and detect any video advertisements using the *EasyList* technology. Upon identifying an ad, the extension localizes and collects the ad metadata, sending this information to the *Server*.

When the *Web User* watches a video ad, the *Browser Extension* tracks user engagement by initiating *WebGazer.js* for gaze tracking, generating heatmaps, and using *DeepFace* for demographic and emotion detection. The resulting gaze data, emotional responses, and demographic insights are tagged with timestamps and sent to the *Server*.

The *Server Module* performs object detection on the ad frames using the *YOLO* model, identifies key objects, and combines the detected objects with the user engagement data. This information is stored for further analysis.

Finally, the *Advertising Company* accesses the *Analytics Frontend* to log in and view the performance metrics of specific ads. The advertiser can request ad-specific data such as gaze heatmaps, demographic information, and emotion metrics. The system generates comprehensive reports that can be downloaded or exported, providing valuable insights for optimizing future ad content and targeting strategies.

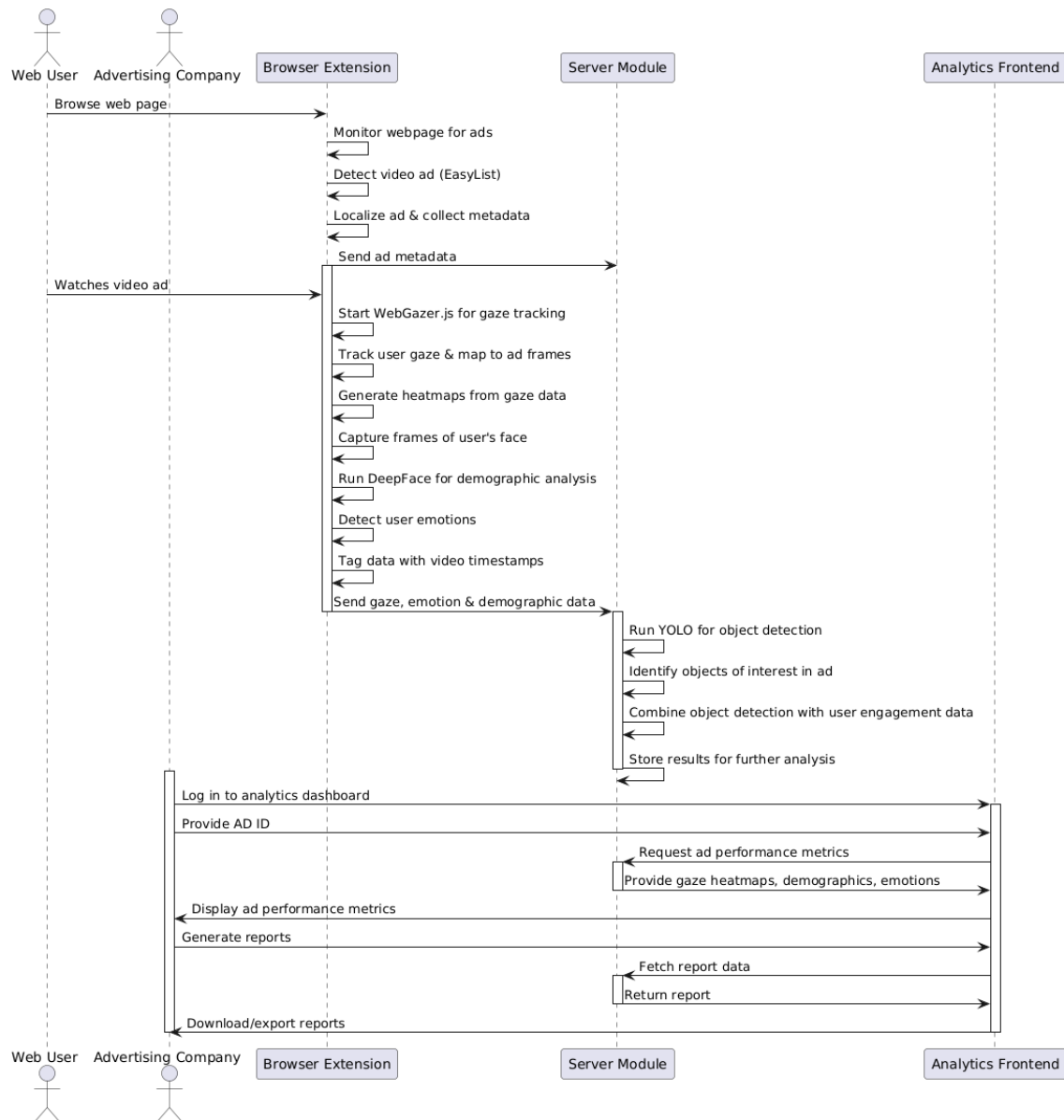


Figure 3.7: System Sequence Diagram

3.3 Data Design

This diagram outlines the structure of the system's data storage. Key entities like Ads, Users, Gaze Data, and Emotions are represented, along with their relationships. For example, each Ad may be linked to multiple Gaze Points and Emotion Records. The diagram highlights how data is organized in the database and how different datasets interact to generate insights for advertisers.

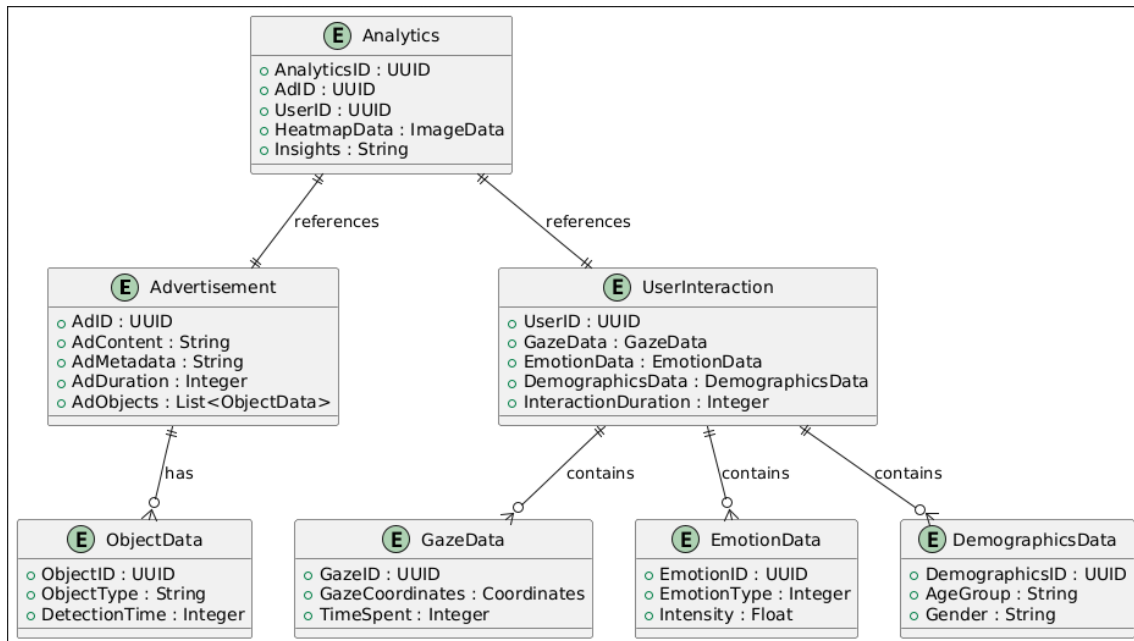


Figure 3.8: Data Representation Diagram

Bibliography

- [1] Abdul Manan Feroz, Abher Ahmed, and Asif Ali. F24-059-D-AdInsight (Report 1). 2024.